7.1 Report Definition

To support the business objective of providing real-time visibility for management established in **Milestone 1**, as well as evaluating operational improvements achieved after replacing the previous spreadsheet-based approach identified in **Milestone 2**, GreenGrid Utilities utilizes Salesforce's built-in reporting capabilities.

The reporting framework is designed using both standard and custom report types that integrate data across multiple related objects. This approach enables the organization to consolidate operational information into meaningful analytical outputs while minimizing duplicate records and maintaining data consistency throughout the CRM environment.

7.1.1 Core Operational Report Specifications

a. Report 1: Average Handling Time (AHT) and Case Triage Performance

- **Report Type:** Cases (Standard Report Type)
- **Format:** Summary Report grouped by **Owner** and **Priority**
- **Key Metrics:**
    - Average Case Age (Hours/Minutes)
    - Total Number of Cases
    - Case Status Distribution
- **Business Purpose:**

This report is designed to monitor the achievement of **Business Requirement BR1**, which targets a **25% reduction in Average Handling Time (AHT)**. By measuring the time required for support representatives to receive, process, and route customer outage requests, management can evaluate service efficiency improvements compared to the previous spreadsheet-driven process. The report also provides insight into workload distribution and case prioritization performance across customer service personnel.

b. Report 2: Field Visit Resolution Efficiency

- **Report Type:** Cases with Field Visits (Custom Report Type utilizing the relationship between Case and Field_Visit__c)
- **Format:** Matrix Report
  - Rows grouped by **Field Technician (Field_Visit__c Owner)**
  - Columns grouped by **Repair Status (Repair_Status__c)**
- **Key Metrics:**
  - Total Dispatch Count
  - Resolution Success Rate
  - Average Service Completion Time
- **Business Purpose:**

This report measures the effectiveness of field operations by evaluating technician performance and repair outcomes. It directly supports **Business Requirement BR3**, which aims to improve first-visit resolution rates by 15 percent. In addition, the report provides visibility into dispatch activities that were previously coordinated through informal communication channels, thereby improving accountability and operational tracking.

c. Report 3: High-Priority Grid Outage Trends

- **Report Type:** Cases with Utility Assets (Custom Report Type utilizing the relationship between Case and Utility_Asset__c)
- **Format:** Summary Report grouped by:
  - Asset Status
  - Installation Address or Service Region
- **Key Metrics:**
  - Total High-Priority Cases
  - Outage Duration Monitoring
  - Asset Failure Frequency
- **Business Purpose:**

This report provides executive management with a comprehensive overview of recurring infrastructure issues. By identifying utility assets that generate repeated service interruptions, leadership teams can prioritize maintenance activities, allocate resources more effectively, and make strategic decisions regarding infrastructure upgrades and reliability improvements.

7.2 Dashboard Design

The dashboard environment serves as a centralized operational monitoring platform for executive leadership. Instead of reviewing raw report records individually, decision-makers can analyze key performance indicators through visual components that provide real-time operational insights.

The dashboard consolidates customer service metrics, field technician activities, and infrastructure performance information into a single management interface, enabling rapid assessment of organizational performance.

7.2.1 Executive Operations Dashboard Blueprint

```
┌──────────────────────────────────────────────────────────────
──────────────────────────┐
│            EXECUTIVE OPERATIONS DASHBOARD           │
├─────────────────────────────┬──────────────────────────────┬───
──────────────────────────┤
│ [Metric Component]  │  [Gauge Component]  │ [Donut Component]  │
│          │          │          │
│  Total Live Open   │ Average Handling Time │  Field Repair Fix  │
│  High-Priority Cases│      (AHT)      │   Success Rates   │
│     -> 14 <-     │  Target: <2 Minutes  │ Resolved vs Failed │
├─────────────────────────────┴──────────────────────────────┴───
──────────────────────────┤
│           [Vertical Bar Chart Component]          │
│                               │
│        Outage Volume & Distribution by Asset Location       │
│           (Region A vs Region B vs Region C)          │
```

Component 1 (Top Left) – Open Crisis Metric

This metric component displays the total number of currently active High-Priority Cases with a status of **New** or **Working**. Because the value is updated automatically through the dispatch automation process implemented in Milestone 6, management can immediately identify ongoing service disruptions requiring attention.

Component 2 (Top Center) – Service Console Speed Gauge

The gauge visualization measures Average Handling Time and compares performance against predefined service targets. The green performance zone represents optimal handling efficiency below two minutes, allowing management to verify that customer support operations have improved beyond the limitations of the former spreadsheet-based workflow.

Component 3 (Top Right) – First-Visit Fix Ratio Donut

This donut chart visualizes the distribution of Field Visit repair outcomes, including Assigned, In Progress, Resolved, and Failed statuses. The component assists management in evaluating technician effectiveness and monitoring field-service productivity.

Component 4 (Bottom Main) – Grid Outage Distribution Bar Chart

The bar chart presents the distribution of outage incidents by installation location or service region. This visualization supports infrastructure planning by identifying geographic areas that experience a higher concentration of utility service disruptions.

7.3 Data Visualization Strategy

To comply with the performance requirements defined in **NFR3**, which specify that dashboards should load within five seconds, GreenGrid Utilities implements structured governance controls for dashboard access, execution context, and data visibility.

These controls ensure that analytical resources remain both performant and secure while supporting organizational decision-making processes.

7.3.1 Role Hierarchy Integration and Dashboard View Context

- Dashboards are configured to execute using a predefined management-level viewing context.
- The "View Dashboard As" setting is assigned to an Executive Manager or designated administrative user.
- This configuration enables Salesforce to process large multi-object datasets efficiently while reducing delays associated with real-time sharing recalculations.
- As a result, dashboard performance remains consistent and aligned with organizational response-time requirements.

7.3.2 Folder Structure and Security Governance

- All reports and dashboards are stored within a dedicated analytics folder named **Executive Operational Analytics**.
- Folder-level permissions are configured according to organizational responsibilities.

Access Permissions

User Group	**Access Level**
Customer Support Team	Viewer
Field Technicians	Viewer
Executive Management	Manager / Editor
IT System Administrators	Manager / Editor

This structure ensures that operational information remains accessible to authorized personnel while preventing unauthorized modifications to executive reporting resources.

7.4 Salesforce Implementation and Configuration Steps

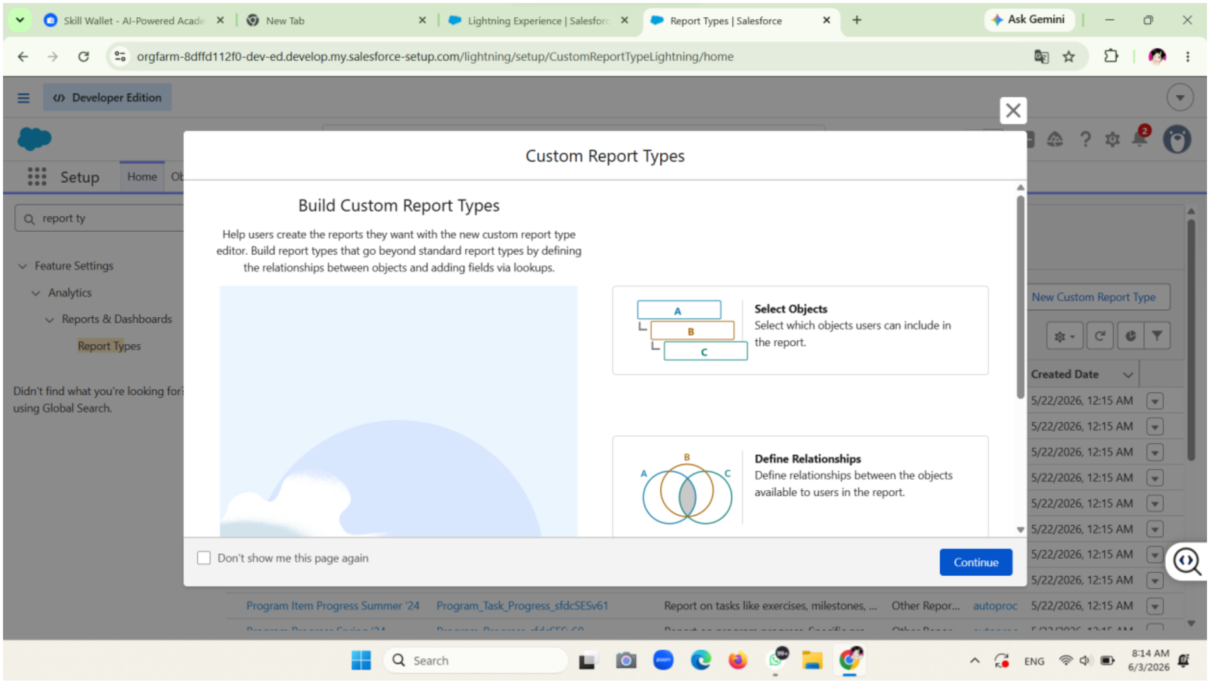
The following section outlines the implementation procedures required to configure reporting and analytics features within the Salesforce environment. These activities include the creation of custom report types, report generation, dashboard construction, and access control configuration. The implementation supports the reporting requirements identified in previous milestones and enables management to monitor operational performance through real-time analytics.

Step 1: Create Custom Report Types for Multi-Object Relationships

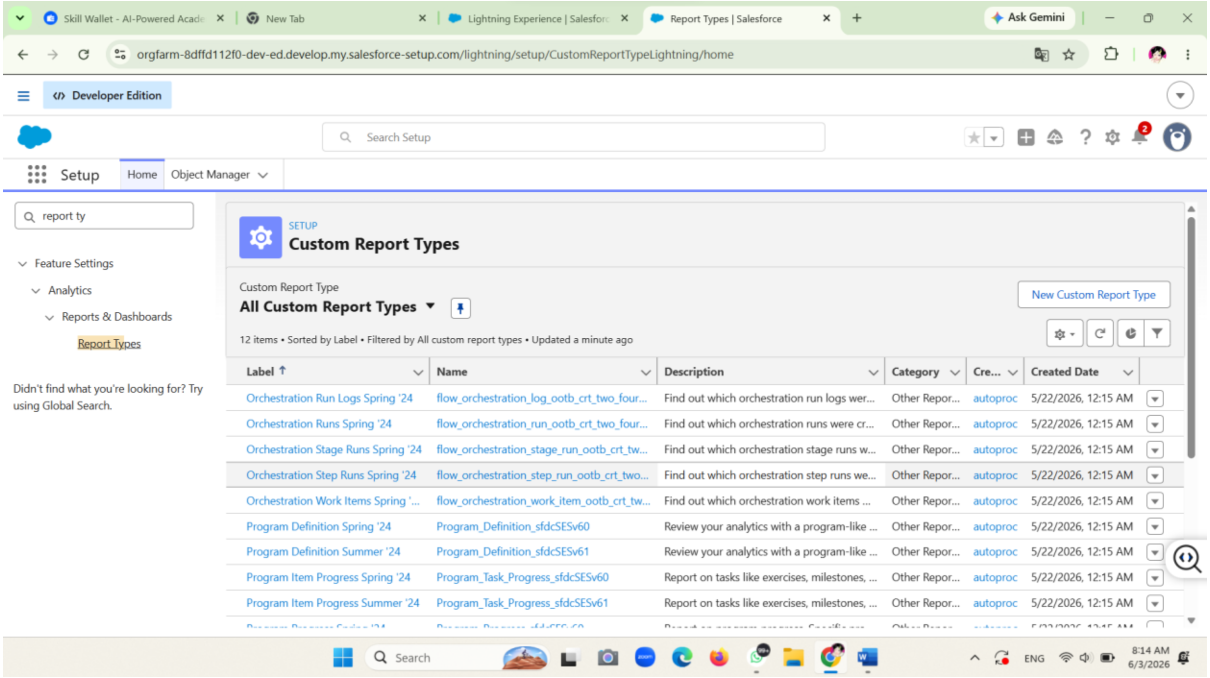
Before developing reports, custom report structures must first be established to connect standard Salesforce objects with the custom objects created in previous milestones, namely **Utility_Asset__c** and **Field_Visit__c**.

Procedure

1. Navigate to **Setup** by clicking the gear icon located in the upper-right corner of Salesforce.
2. In the **Quick Find** search field, type **Report Types** and select **Custom Report Types**.

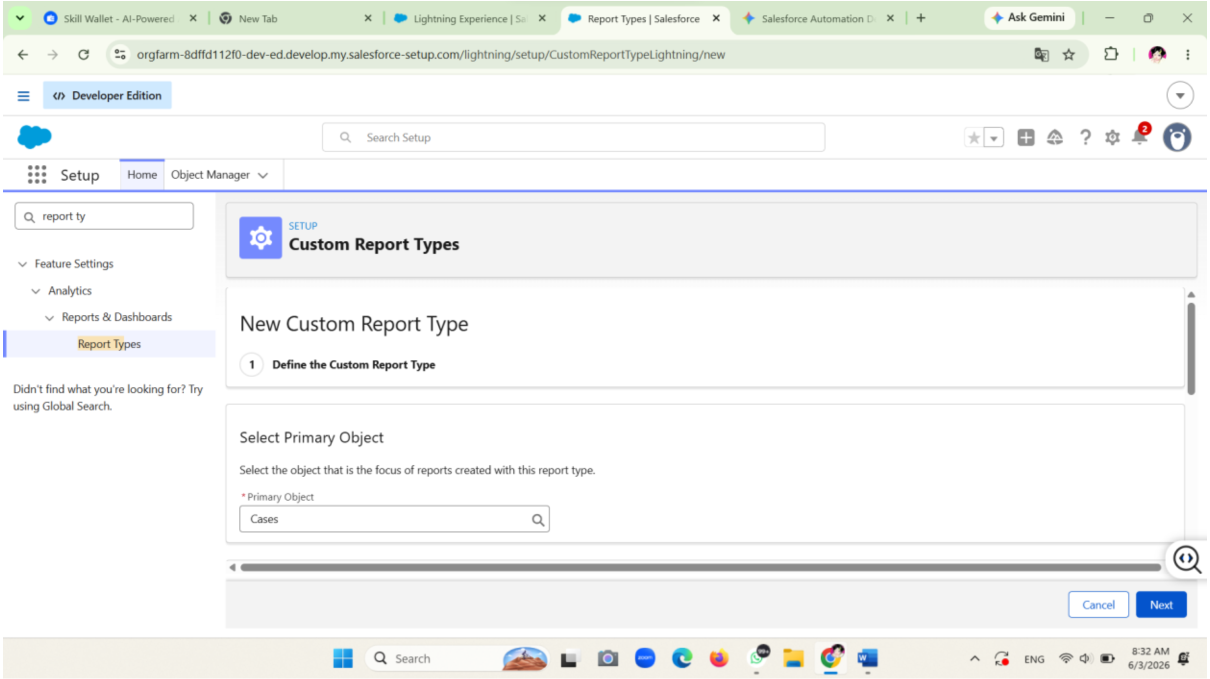


3. If an introductory page appears, click **Continue**.
4. Click the **New Custom Report Type** button.



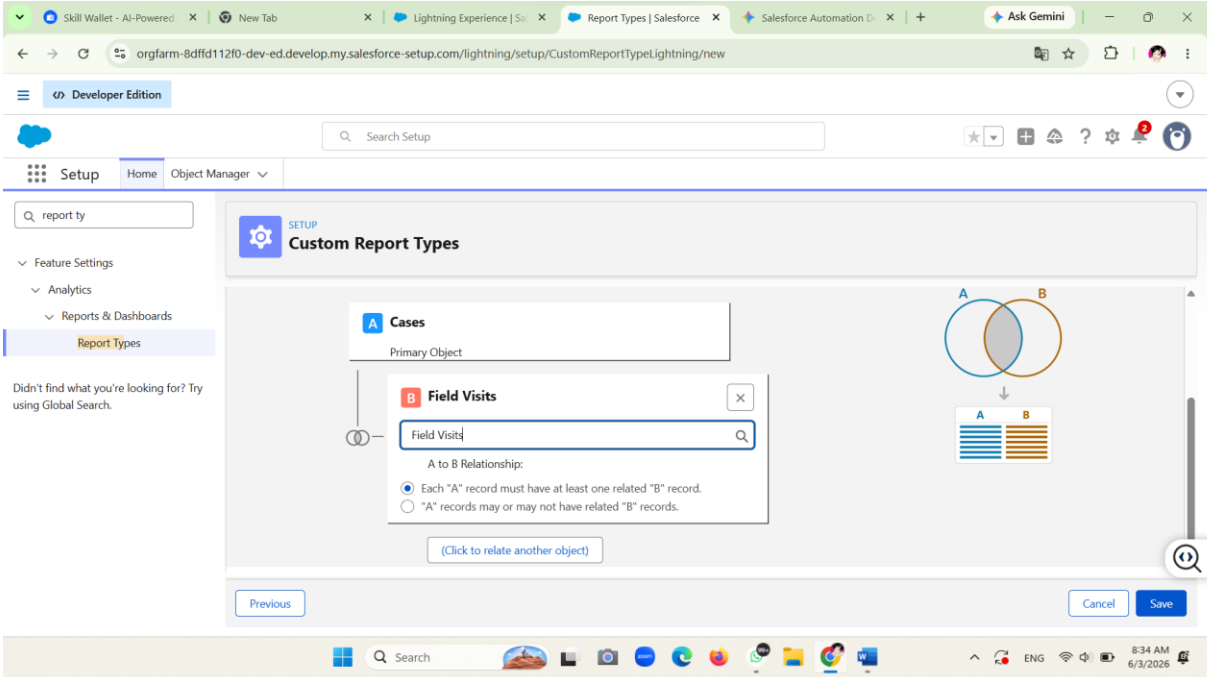
5. Configure the report type settings as follows:

Configuration Item	**Value**
Primary Object	Cases
Report Type Label	Cases with Field Visits
Report Type Name	Cases_with_Field_Visits
Description	Operational report linking customer service tickets to field crew dispatch actions
Store in Category	Customer Support Cases
Deployment Status	Deployed



6. Click **Next**.
7. On the **Define Object Relationships** page, select the empty box located below **A (Cases)** to add a related child object.
8. Choose **Field Visits (Field_Visit__c)** from the available list. Retain the default relationship setting indicating that each Case record must have at least one related Field Visit record.
9. Click **Save**.
10. To create an additional report type for asset analysis, repeat the process by clicking **New Custom Report Type** again and configure the following:

Configuration Item	Value
Primary Object	Cases
Related Object	Utility Assets (Utility_Asset__c)
Report Type Label	Cases with Utility Assets
Deployment Status	Deployed



11. Click **Save** to complete the configuration.

Expected Result

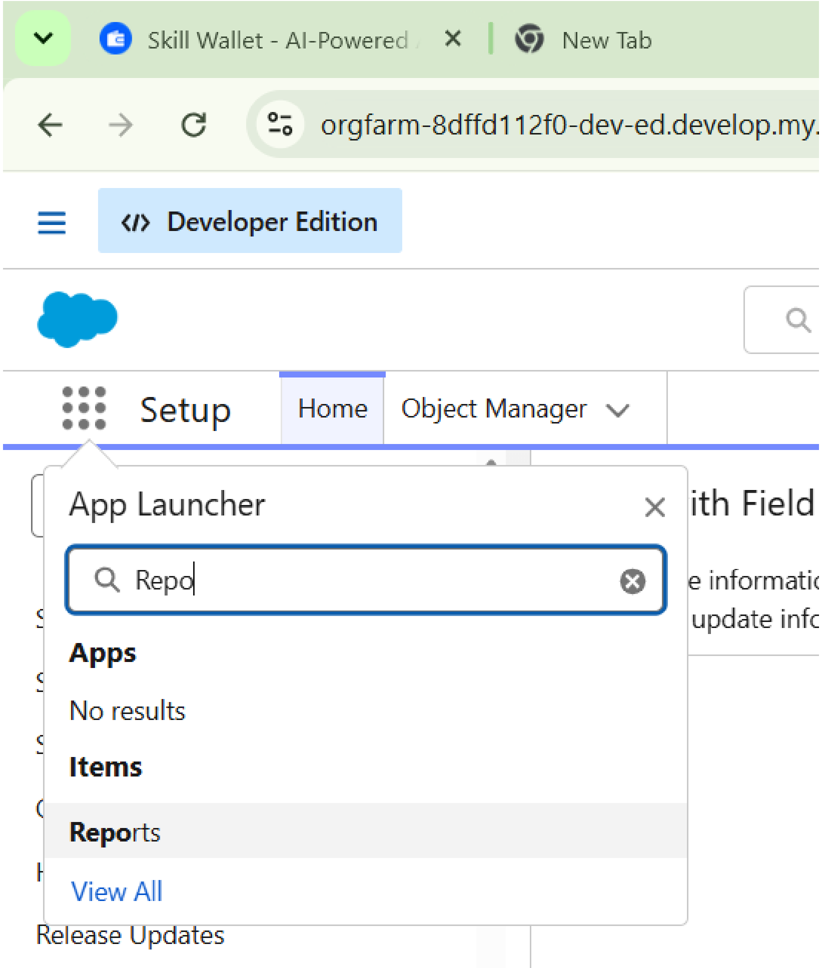
The system successfully creates custom report types that combine Case records with Field Visit and Utility Asset information, enabling multi-object reporting capabilities.

Step 2: Build Report 1 — Average Handling Time (AHT) Analytics

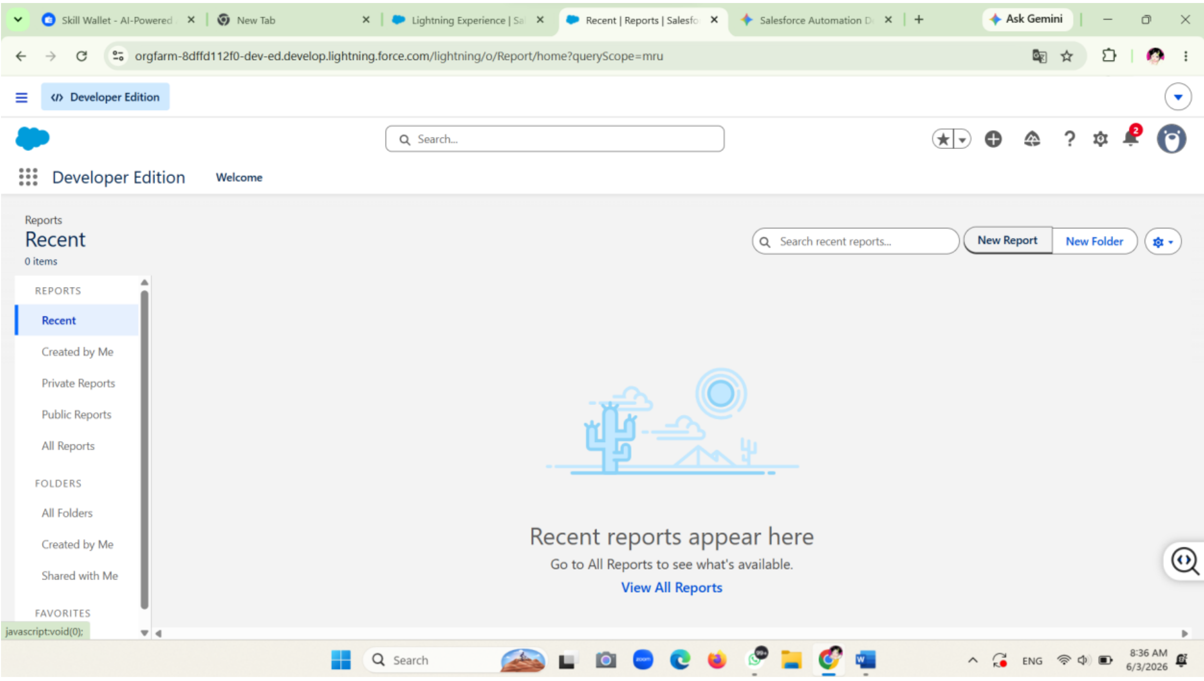
This report is designed to evaluate customer support performance by measuring how efficiently service requests are processed and managed.

Procedure

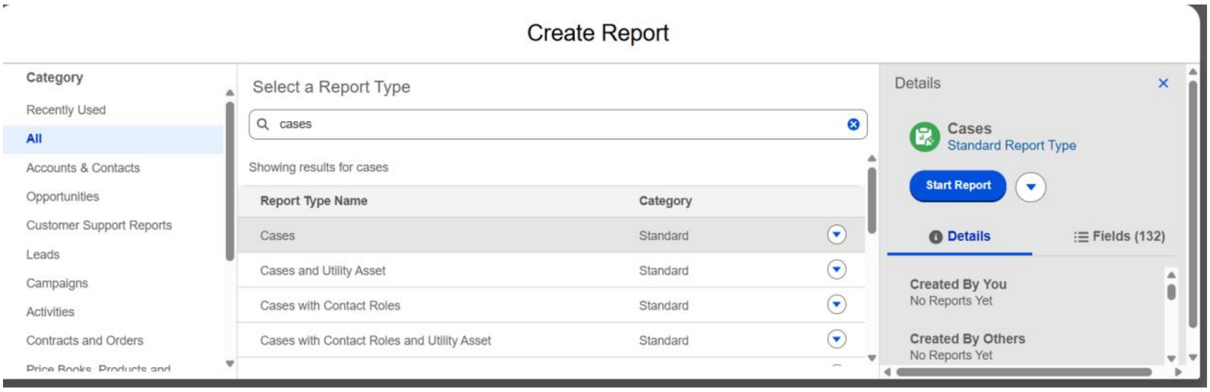
1. Open the **App Launcher** and search for **Reports**.



2. Select **Reports**.

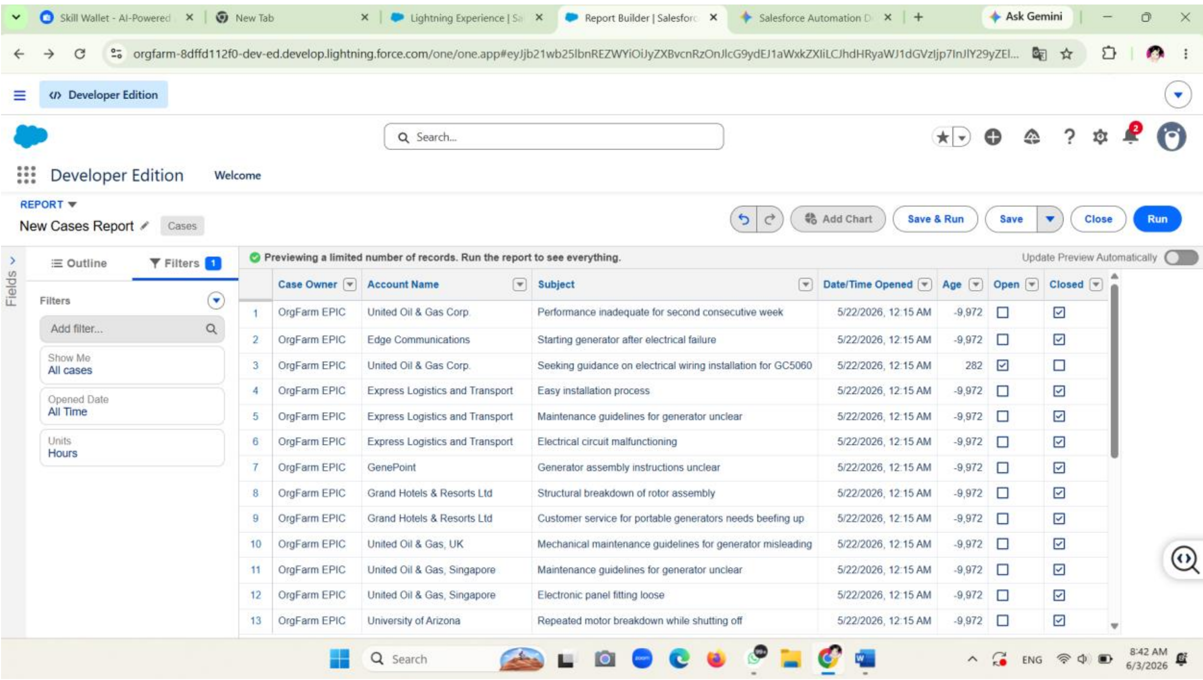


3. Click **New Report**.
4. Search for and choose the standard **Cases** report type.
5. Click **Start Report**.



6. Enable **Update Preview Automatically**.
7. Configure the report filters:

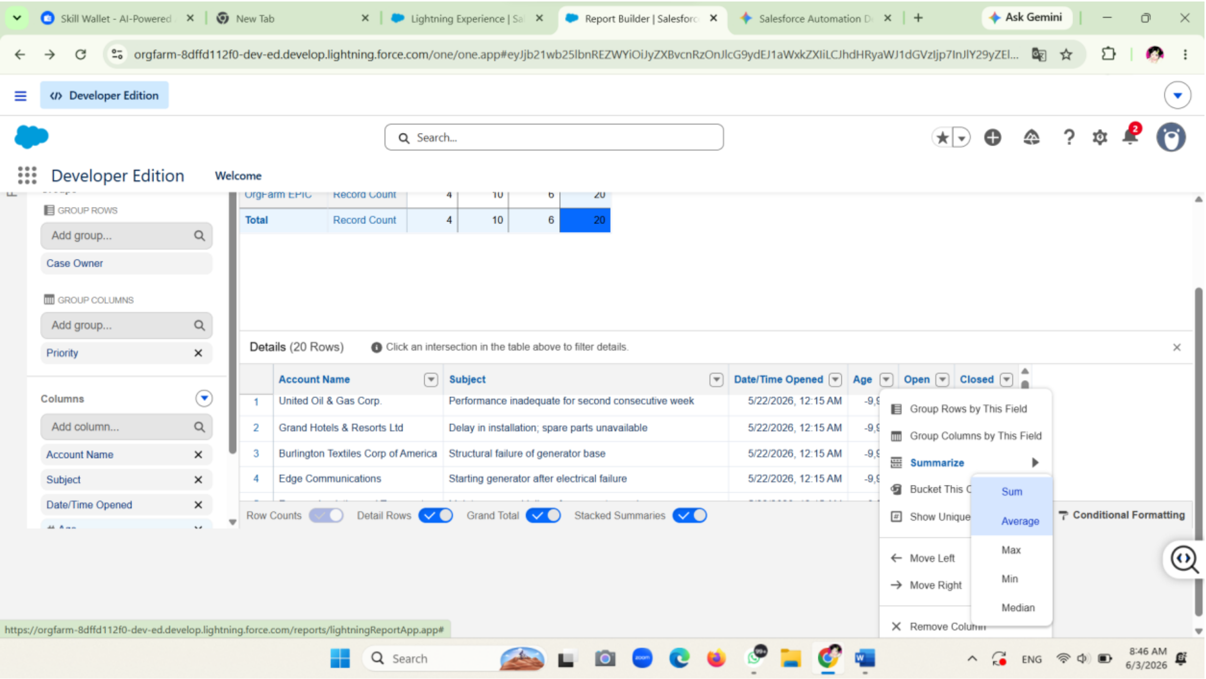
Filter	**Value**
Show Me	All Cases
Opened Date	Current FQ (or All Time for testing purposes)



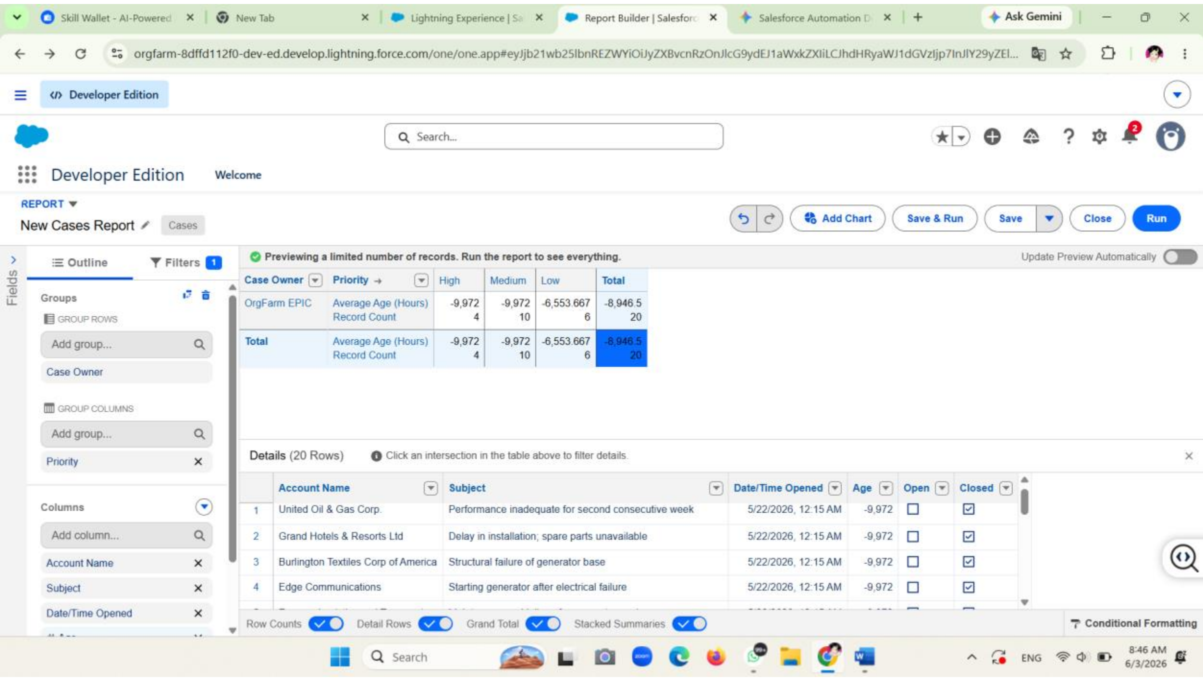
8. Under the **Outline** section:
   - Group rows by **Case Owner**.

- Add a second grouping by **Priority**.

9. Add the **Age** column to the report layout and remove unnecessary fields.
10. Apply the **Average** summary function to the Age field.



11. Click **Save & Run**.



12. Save the report using the following details:

Setting	Value
Report Name	AHT and Case Triage Performance
Unique Name	AHT_and_Case_Triage_Performance

Expected Result

The report displays average handling time metrics, case volumes, and priority-based performance information for customer support operations.

Step 3: Build Report 2 — Field Visit Resolution Efficiency

This report measures technician performance and field service effectiveness by tracking dispatch activities and repair outcomes.

Procedure

1. Click **New Report**.
2. Select **Cases with Field Visits**.
3. Click **Start Report**.
4. Configure the filters:

Filter	Value
Show Me	All Cases
Opened Date	All Time

5. Configure the report outline:
   - Group Rows by **Field Visit Owner**.
   - Group Columns by **Repair Status**.
6. Add the following fields:
   - Visit Number
   - Technician Notes
7. Click **Save & Run**.
8. Save the report using:

Setting	Value
Report Name	Field Visit Resolution Efficiency

Unique Name	Field_Visit_Resolution_Efficiency

Expected Result

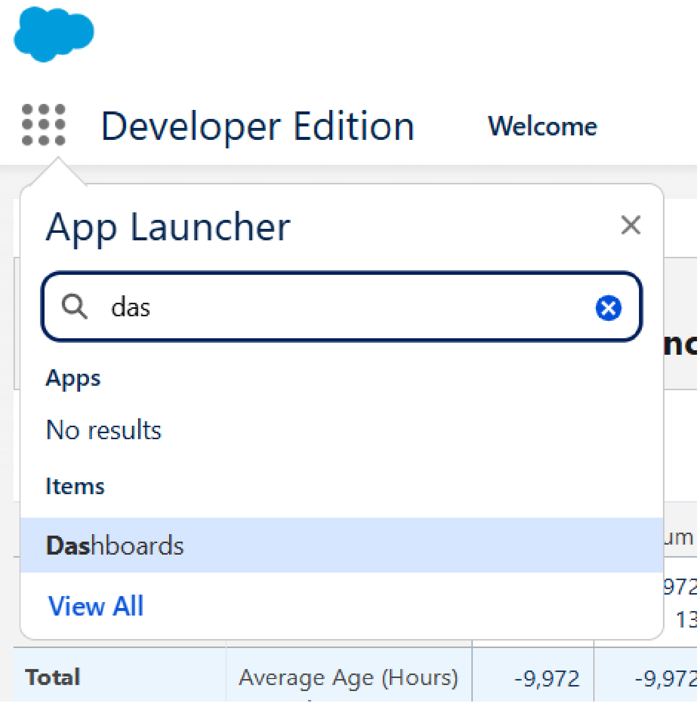
The report presents repair completion statistics, technician activities, and resolution outcomes for field operations.

Step 4: Provision Secured Analytics Folders

To improve report organization and maintain access control, a dedicated analytics folder must be created.

Procedure

1. Open the **Dashboards** tab.



2. Click **New Dashboard Folder**.
3. Enter the folder name:
    - **Executive Operational Analytics**
4. Click **Save**.
5. Open the **Share** settings for the folder.
6. Configure access permissions as follows:

User Group	**Access Level**
Executive Management	Manage
Customer Support Team	Viewer

7. Click **Done**.

Expected Result

A secured dashboard folder is created and appropriate access permissions are assigned based on organizational roles.

Step 5: Assemble the Executive Operations Dashboard

The dashboard serves as a centralized monitoring interface for management and operational leadership.

Procedure

1. Open the **Dashboards** tab.
2. Click **New Dashboard**.
3. Configure the dashboard information:

Configuration Item	**Value**
Name	Executive Operations Dashboard
Description	Real-time grid management overview monitoring call triage speed, asset status, and mobile technician field repair rates
Folder	Executive Operational Analytics

4. Click **Create**.

Step 6: Add Component 1 — Live Open Crisis Counter

This component displays the number of active high-priority service incidents currently being processed.

Procedure

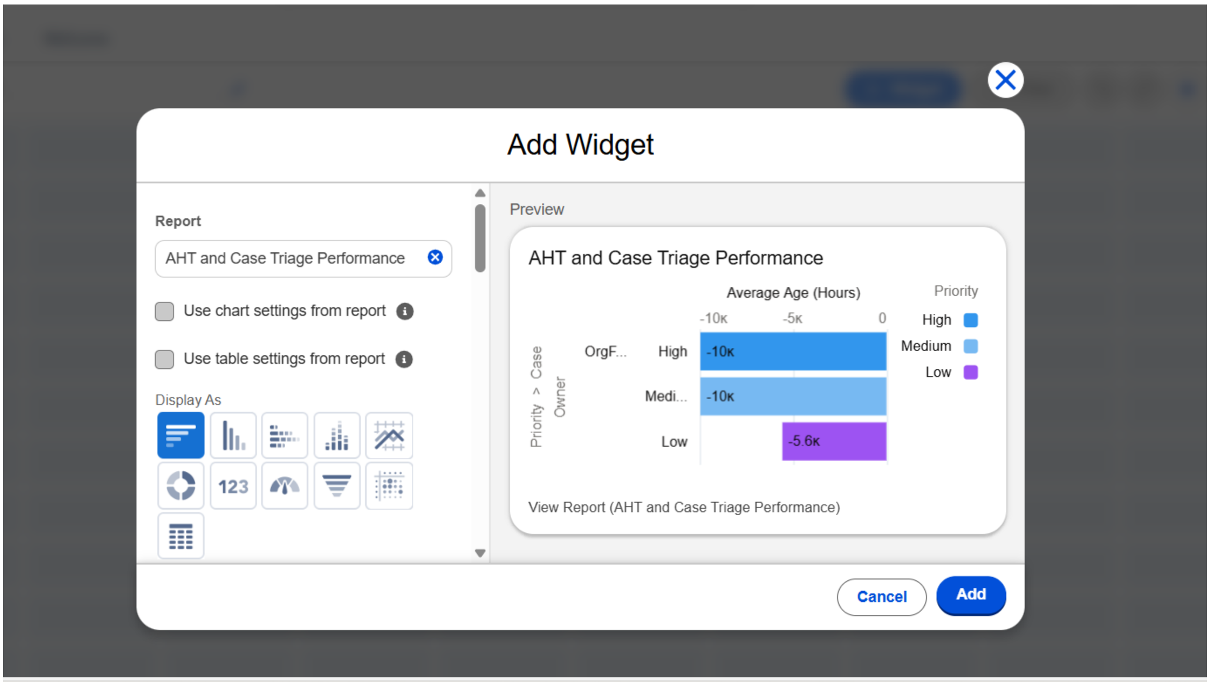
1. Click **+ Component**.
2. Select the report:
    - **AHT and Case Triage Performance**
3. Configure the component settings:

Configuration	**Value**
Component Type	Metric
Measure	Record Count
Title	Live Open High-Priority Cases

4. Click **Add** and place the component in the upper-left section of the dashboard.

Expected Result

The dashboard displays a real-time count of active high-priority cases.



Step 7: Add Component 2 — Service Console Speed Gauge

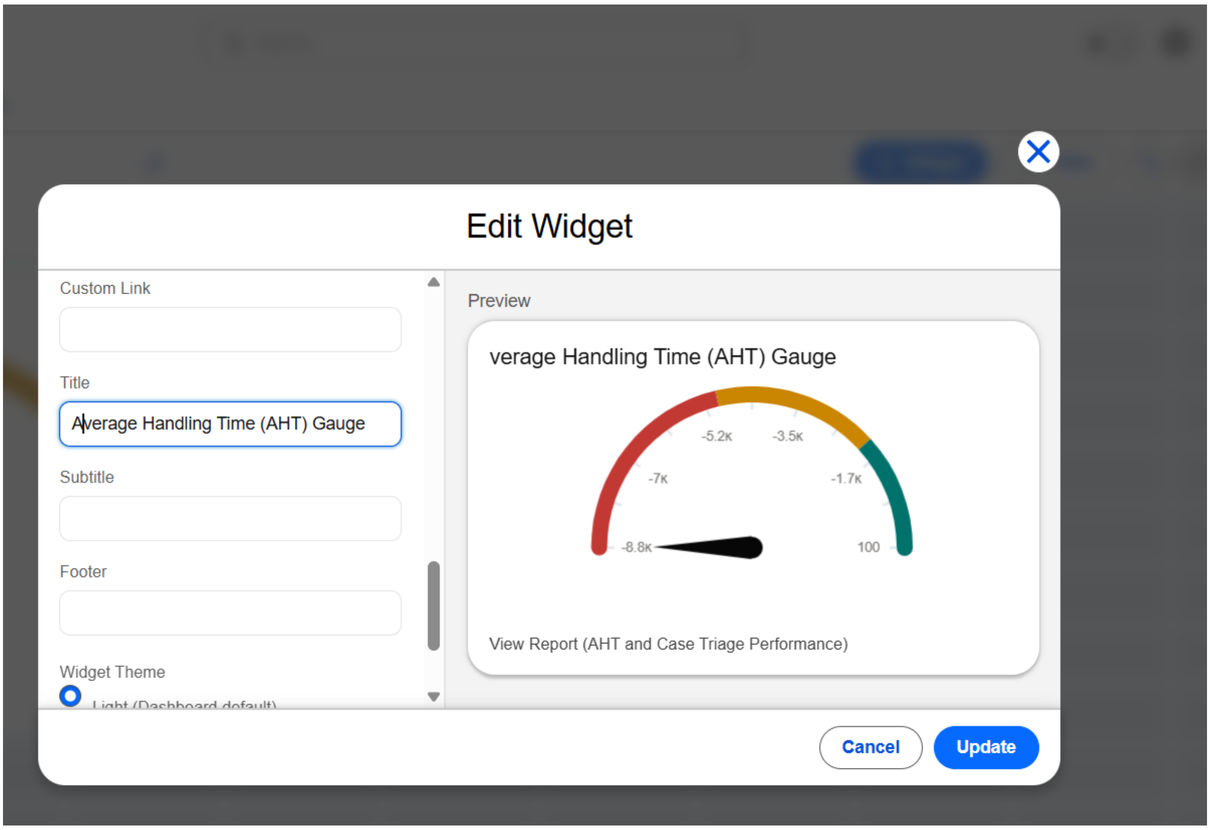
This component visualizes Average Handling Time performance against predefined service targets.

Procedure

1. Click + **Component**.
2. Select:
   - **AHT and Case Triage Performance**
3. Configure the component:

Configuration	Value
Component Type	Gauge
Measure	Average Age
Breakpoint 1	2
Breakpoint 2	5
Title	Average Handling Time (AHT) Gauge

4. Click **Add** and position the component at the top-center section of the dashboard.



Step 8: Add Component 3 — First-Visit Fix Ratio Donut

This component visualizes repair completion outcomes and technician effectiveness.

Procedure

1. Click + **Component**.
2. Select:
    - **Field Visit Resolution Efficiency**
3. Configure:

Configuration	Value
Component Type	Donut Chart
Metric	Record Count
Slice By	Repair Status
Title	Field Repair Fix Success Rates

4. Click **Add** and position the chart in the upper-right section of the dashboard.

Expected Result

The donut chart displays the distribution of repair statuses and first-visit resolution performance.